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Metagenomics Solutions

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Metagenome Human Gut V3-V4 MG-HGAS-VTF

Applications

Method utilizes targeted amplification and next-generation sequencing of the 16S rRNA gene V3–V4 hyper-variable regions to profile bacterial communities in human gut samples. This applied genomics approach enables accurate taxonomic characterization and comparative analysis of gut microbiome composition across biological conditions. The V3–V4 metagenome is a cost-effective and scalable method, where the primary objective is to measure changes in microbial diversity and analyze beta-diversity. These methods are ideal for characterized metagenomes with well-documented reference datasets, where the primary objective is understanding treatment differences within a cohort. The method is limited in resolving species- and genus-level taxonomy.

Pros

- High-resolution profiling of gut bacteria
- Cost-effective microbiome analysis
- Suitable for large cohort studies
- Robust taxonomic classification
- deal for comparative gut microbiome studies

Notes

- Limited to bacterial community profiling
- Species-level resolution may vary on match from V3-V4
- Functional insights are predictive
- Results influenced by sample handling and storage

Input

DNA ≥ 10 ng recommended

DNA Qubit > 10ng

Sample Types

Stool samples

Swabs

*Stabilized samples | Fresh/Frozen not accepted

Platform

Illumina platforms | 0.1 - 0.2M | 300 x 2

Process Map

Sample Extraction > Sample QC > 16S Amplicon QC & Library QC > NGS Data Acquisition > Primary Data QC > Data Clensing ASV > QIIME + RDP > Taxonomic Profile > β diversity

Standard Deliverables

Download Sample Report

- Sample & Library QC and summary
- Method & Workflow Summary
- Species-level diversity and abundance analysis using SILVA, Green-genes, and 16S-NCBI reference databases
- Read Clustering & Rarefaction Analysis
- Species/Genus Identification - Shannon, Simpson, and Chao diversity index
- PCA /PCoA plots
- Differnetial- Heatmaps
- KRONA Charts & Predicted Microbial Pathways (optional)

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Metagenome Human Gut 16s Whole Amplicon MG-HGAS-WHO

Applications

Method utilizes full-length 16S rRNA gene (full amplicon) sequencing to profile bacterial communities and improve taxonomic resolution. By capturing the complete 16S gene, the approach enables more accurate genus- and species-level classification compared to partial-fragment sequencing. Compared to the V3–V4 or other fragment-based methods, long amplicon 16S reads are matches bacterial species and genus. The data provide higher confidence in the discovery of species and inference of microbial pathways, leading to a superior estimate of alpha diversity. This method can be applied to well-referenced metagenomes such as the human gut, as well as environmental samples.

Pros

- Superior resolution 16S for diversity analysis
- Enhanced species-level identification
- Reduced primer-region bias
- Ideal for detailed microbiome profiling
- Suitable for cohort and longitudinal studies species resolution is desired

Notes

- Focused on bacterial community analysis
- Functional insights are predictive, not direct
- Species-level resolution depends on reference databases
- Sensitive to DNA quality and extraction/amplification

Input

Microbial DNA ≥ 10 ng recommended

DNA Qubit > 10ng

Sample Types

Stool samples | Swabs | Gut-derived clinical samples

Platform

ONT sequencing platforms | Long reads | 50K - 100K

Process Map

Standard Deliverables

Download Sample Report

- Sample & Library QC
- Method & Workflow Summary
- Raw Data QC & Processing
- Read Clustering, Matching, and Rarefaction Analysis
- Species/Genus Identification with Shannon/ Simpson Indices
- Differential Taxonomic Abundance with Heatmaps
- KRONA Charts & Predicted Microbial Pathways (optional)

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Integrated Microbiome Analysis – Fungal & Bacterial

MG-INTM-FBC

Applications

Method utilizes integrated amplicon sequencing of bacterial and fungal marker genes to comprehensively profile mixed microbial communities within a single study. Bacterial communities are characterized using 16S rRNA gene sequencing, while fungal communities are profiled using ITS (ITS1/ITS2) or 18S rRNA regions. This integrated approach enables simultaneous assessment of bacterial–fungal composition, diversity, and co-occurrence patterns across biological or environmental conditions. The use of long amplicons provides the ability to type microbial as well as fungal/plant samples present in a microbiome. These investigations are well suited to profile species richness, diversity, and abundances in microbiomes to infer changes in time-course or treatment-dependent studies. Inference of alterations in microbial abundances provides clues to shifts in microbial pathways.
These methods are suitable when host DNA is high, sample DNA is limiting, or as a pre-pilot for whole-genome, metagenome, or metatranscriptome investigations.

Pros

- Simultaneous profiling of bacterial and fungal communities
- Enables cross-kingdom interaction analysis
- Improved ecological and functional interpretation
- Suitable for complex microbiome studies

Notes

- Based on targeted marker gene sequencing
- Functional insights are predictive
- Taxonomic resolution depends on reference databases
- Sensitive to sample quality and inhibitors
- ITS- may not detect all fungal types with equal efficiency

Input

Microbial / Environmental DNA
≥ 10 ng recommended

DNA Qubit > 10ng

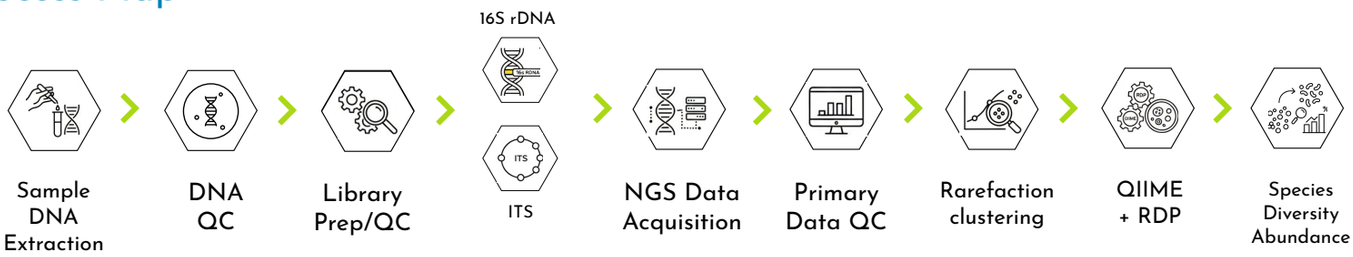
Sample Types

Infected Tissues | Soil | Extreme environment samples | Biofilms and sludge | Water

Platform

ONT Long-read sequencing | 200K-250K reads 16s | 50K reads ITS

Process Map



Standard Deliverables

Download Sample Report

- Sample QC, Library QC, Summary of methods
- Data QC / Sequence data preprocessing summary
- 16S species match for typing and abundances
- ITS match for fungal and abundances
- Rarefaction Plots
- Species summary and counts
- Measure of diversity with Shannon & Simpson Indices
- When multiple samples in time covers / treatment are provided in Beta diversity, differential heatmaps and PCA plots provided

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Whole Genome Sequencing – Microbe (De novo) AG-WGMI-DNV

Applications

Method utilizes shotgun whole-genome sequencing to generate a de novo assembly of microbial genomes without reliance on a reference sequence. This approach enables complete characterization of bacterial, archaeal, and fungal genomes, including novel species or strains. De novo assembly allows identification of coding and non-coding regions, genomic rearrangements, plasmids, and mobile genetic elements.

Pros

- Enables discovery of novel genomes and genes
- High-resolution strain characterization
- Suitable for poorly characterized organisms
- Supports functional and evolutionary studies
- Identifies Episomes, Virions etc
- Base modification in DNA

Notes

- Requires high-quality genomic DNA
- Assembly quality depends on sequencing depth
- Complex genomes may require long-read support
- 16S/18S purity testing suggested

Input

High Molecular Genomic DNA

- Short-read sequencing: ≥ 400ng - 1µg
- Long-read sequencing (if applicable): ≥ 1µg
- DIN/Gel analysis suggested
- 1 - 10ng low input prep will not allow DNA base modification calling

Sample Types

Bacterial isolates | Archaeal isolates | Fungal isolates

Platform

Illumina platforms / Long-read sequencing platforms | 100X Depth | 30X - 40X depth on long read

Process Map

Genomic DNA Extraction > Sample QC > NGS Library > NGS Data (Longread) > Data QC > Sequence Assembly > Gene Discovery Annotation > Genome Annotation

Standard Deliverables

Download Sample Report

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, and pre-processing
- Genome Assembly Summary Statistics,
- Gene Calling, Gene Annotation
- Identification of Transposable Elements and Gene Clusters
- Antibiotic / Resistance Loci
- GFF3 File, Gene List, Annotation–Pathway Assignment, Gene Ontology Assignment
- CIRCOS Plots, Genome Maps

*Where Applicable: Episome Assembly and Annotation

Additional option

- DNA methylation analysis

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Whole Genome Sequencing – Fungal (De novo) AG-WGFU-DNV

Applications

Method utilizes shotgun whole-genome sequencing to generate a de novo assembly of fungal genomes without reliance on a reference sequence. This enables comprehensive characterization of fungal genomes, including novel species or strains, complex gene families, repetitive regions, and secondary metabolite biosynthetic clusters. De novo assembly supports studies in fungal biology, pathogenicity, industrial strain development, and evolutionary genomics.

Pros

- Enables discovery of novel fungal genes
- Captures complex and repetitive genomic regions
- Suitable for poorly characterized fungal species
- Supports functional and evolutionary studies
- Structural variation, synteny and genome organization
- Repeat/Recombination events
- Haplotype phasing

Notes

- Requires high-quality genomic DNA of high molecular weight
- Assembly quality depends on genome size and complexity
- Long-read sequencing recommended for repeat-rich genomes
- 18S strain purity testing recommended

Input

Genomic DNA

- Short-read sequencing: ≥ 200 ng
- Long-read sequencing (recommended): ≥ 1 µg
- DIN/Gel analysis for high molecular weight DNA

Sample Types

- Fungal isolates
- Clinical fungal isolates
- Environmental fungal isolates
- Industrial and agricultural fungal strains

Platform

Long-read sequencing platforms | 50-70X Long read | ~ 2-10 GB | Based on genome/ploidy estimates

Process Map

Genomic DNA Extraction > Sample QC > NGS Library > NGS Data (Longread) > Data QC > Genome Assembly > Gene Discovery Annotation > Genome Analysis

Standard Deliverables

Download Sample Report

- Sample QC, Library QC, and Summary of Methods
- Data QC, Data Trimming, and pre-processing
- Genome Assembly Summary, Gene Detection, Genome Annotation
- SNP/Allele identification
- Haplotype information from long read parsing
- GFF3 File, Gene List, Repeat Elements, SINE/LINES, Pathway assignment, Secondary metabolite coding genes
- When multiple Strain/Strain matching scoped - synteny and comparitive genome mapping for specific project types

References

- Estaki M et al. Curr Protoc Bioinformatics. 2020; PMID:32343490.
- Licata AG et al. Microbiol Spectr. 2025; doi:10.1128/spectrum.01673-25.

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